CHM-DSM-401 — Fundamental of Chemistry-IV

Unit III: History of Chemistry

7th Semester, Assam University (FYUG) — Haflong Government College

Paper: CHM-DSM-401, Full Marks 100 [ESE 70 + Internal 30] | Pass marks 40 | Unit III: 9 lectures

Notes 18 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

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Unit syllabus (as in the syllabus file)

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7. Discoveries and Inventions in the context of state of art and impact.
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Top 2 high-mark topics in this unit

1. Ayurvedic Chemistry and Alchemy in India — two named traditions, content-heavy descriptive long answer, the most distinctive named topics in the unit.
2. Metal and Metallurgy, and Fibre, cloth, Paper, ink, and dyeing chemistry of ancient India — concrete processes and products, an easy long descriptive answer with examples.

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1. Old Traditions of Chemical Sciences in India

Chemistry as a Practical Science in Ancient India and the Vedic Period

In ancient India, chemistry developed primarily as an applied, practical science, deeply intertwined with medicine, industrial crafts, metallurgy, and religious rituals. Rather than existing as an isolated theoretical discipline, chemical knowledge was cultivated through empirical observation, experimentation, and systematic refining of techniques across generations. The early chemical tradition was commonly known in broad contexts under terms such as *Rasayana-shastra* (the science of chemical transformations and alchemy) and *Vaidyaka-shastra* (pharmaceutical chemistry associated with medicine).

The earliest textual records of chemical operations in India are found in the Vedic literature, dating from the second millennium BCE. The *Rigveda*, *Yajurveda*, and *Atharvaveda* contain explicit references to various metals, extraction methods, dyes, fermented beverages, and crude distillation processes:

1. **Metals and Metallurgy**: The Vedic texts mention several metals under specific terminology. The term *Ayas*, which initially referred to copper or bronze, later expanded to denote iron (*Krishna-ayas* or black metal, as distinguished from *Lohita-ayas* or red metal/copper). Gold (*Hiranya*), silver (*Rajata*), tin (*Trapu*), and lead (*Sisa*) were widely recognized, weighed, and processed for ornaments, coins, and domestic vessels.
2. **Dyes and Pigments**: Textile dyeing was an advanced craft during the Vedic era. Natural dyes were extracted from plant sources such as madder (*Manjistha*), turmeric (*Haridra*), and indigo (*Nili*). Mordants such as alum (*Saurashtri*) were utilized to fix plant dyes permanently to cotton and wool fibers.
3. **Fermentation and Distillation**: The preparation of fermented beverages held both ritualistic and practical significance. The *Rigveda* provides extensive accounts of the preparation of *Soma*, an intoxicating or medicinal plant extract, and *Sura*, a fermented liquor akin to beer or ale. The process of preparing *Sura* involved mash fermentation of barley or rice meal mixed with natural yeast starters (*Kina*), followed by filtration through sheep's wool. Primitive distillation techniques were developed to concentrate spirits and essential oils, utilizing pottery apparatus designed to condense vapors on cooled inverted lids (*Putiyantra*).

Use of Minerals, Salts, Alkalis, and Acids in Daily Life and Crafts

Ancient Indian artisans and practitioners acquired sophisticated empirical knowledge regarding the chemical behavior of naturally occurring minerals, salts, alkalis, and acids, employing them across diverse domestic applications and industrial crafts.

Minerals and Salts

Naturally occurring salts were categorized based on their geographic origins and chemical characteristics. Textual sources identify several distinct salts: * *Saindhava*: Rock salt obtained primarily from the Salt Range, consisting largely of sodium chloride (NaCl) with minor mineral impurities. * *Samudra*: Sea salt obtained through the solar evaporation of seawater. * *Sauvarchala*: Black salt, rich in sodium chloride along with trace sulfur compounds ($\mathrm{Na_2S}$, $\mathrm{H_2S}$) giving it a characteristic odor and flavor. * *Romaka* and *Audbhida*: Saline efflorescences harvested from dried lake beds and alkaline soils.

Minerals such as green vitriol or ferrous sulfate (*Kasisa*), blue vitriol or copper sulfate (*Tuttha*), cinnabar or mercuric sulfide (*Hingula*), realgar (*Manahshila*), and orpiment (*Haritala*) were purified and incorporated into pigments, glass coloring, and medicinal preparations.

Alkalis (*Kshara*)

Alkalis played a central role in ancient chemical operations, serving as cleansers, fluxes in metallurgy, and reagents in medicine. Plant ashes were the primary source of soluble alkalis. Ashes obtained from burning plants such as barley straw (*Yava*), plantain stems, and *Salicornia* species were extracted with water to yield pot-ash (potassium carbonate, $\mathrm{K_2CO_3}$) and soda-ash (sodium carbonate, $\mathrm{Na_2CO_3}$). These were designated as *Yavakshara* and *Svarjikakshara*, respectively.

To increase the chemical strength of mild alkalis, ancient chemists performed caustification by reacting aqueous alkali carbonates with slaked lime derived from roasted limestone or calcined seashells.

Acids (*Amla*)

While strong, isolated mineral acids like concentrated sulfuric or nitric acid were not produced in bulk until later medieval periods, acidic vegetable extracts and acidic salts were widely utilized. Juices of acidic fruits such as tamarind, citrus (*Nimbu*), and fermented acidic gruels (*Kanjika*) provided organic acids such as citric acid, tartaric acid, and acetic acid (*Shukta*). These acidic media were employed to clean metal surfaces, convert metal sheets into fine oxides, dissolve calcined salts, and act as gentle solvents in pharmaceutical preparations.

The Idea of Matter and its Transformation: The Panchamahabhutas and Vaisheshika Atomic Theory

Theoretical frameworks explaining the nature of matter, its constituent particles, and chemical transformations were developed in ancient Indian philosophical schools, notably the *Sankhya* and *Vaisheshika* systems.

The Five Fundamental Elements (*Panchamahabhutas*)

In Indian cosmological and physical thought, all material substances in the universe are composed of five fundamental principles or elements, known as the *Panchamahabhutas*:

1. *Prithvi* (Earth): Associated with the state of solidity, possessing mass, density, rigidity, and the property of smell.
2. *Ap* (Water): Associated with the state of liquidity, possessing fluidity, cohesion, viscosity, and the property of taste.
3. *Tejas* or *Agni* (Fire): Associated with energy, heat, light, and the driving force behind all chemical transformations and changes of state.
4. *Vayu* (Air): Associated with the gaseous state, mobility, pressure, expansion, and touch.
5. *Akasha* (Space/Ether): Associated with volume, non-resistance, sound transmission, and the spatial medium in which matter exists.

Physical matter was understood not as a single element in isolation, but as a combination of all five elements in varying proportions, with one or two elements dominating to determine the physical state and chemical properties of the substance.

Atomic Theory of Kanada and the Vaisheshika School (*Paramanuvada*)

Formulated by Acharya Kanada (c. 6th–2nd century BCE), the *Vaisheshika* school developed an atomic theory of matter (*Paramanuvada*) that predates modern European atomic concepts by several centuries.

According to Kanada, matter cannot be divided infinitely. Continuous division of a physical body leads ultimately to an indivisible, ultimate, spherical particle called a *Paramanu* (atom). A *Paramanu* is eternal, indestructible, and possesses specific qualitative characteristics depending on the element (*Prithvi*, *Ap*, *Tejas*, or *Vayu*) to which it belongs. *Akasha* is considered continuous and non-atomic.

Paramanus in their isolated state are imperceptible to human senses and cannot exist independently in macroscopic matter. To form tangible matter, ultimate atoms must combine in fixed numerical ratios:

First, two ultimate indivisible atoms (*Paramanus*) of identical or compatible nature combine under the influence of an inherent force (*Adrishta* or kinetic impulse) to form a binary atom called a *Dvyanuka* (dyad). Let P represent a single *Paramanu*. The combination forming a *Dyanuka* (D) is expressed as:

$$2P \longrightarrow D$$

Next, three binary atoms (*Dyanukas*) combine together to form a ternary atom, known as a *Tryanuka* or *Trasarenu* (triad). A *Tryanuka* represents the smallest unit of matter that possesses physical dimension and is visible as a speck in a shaft of sunlight. The combination of three dyads (D) to form a triad (T) is given by:

$$3D \longrightarrow T$$

To express the total number of primary atoms (*Paramanus*) contained within a single visible triad (*Tryanuka*), we substitute the expression for D into the formation equation for T:

$$T = 3(2P)$$

Evaluating the product yields the exact atomic composition of a triad:

$$T = 6P$$

Thus, one fundamental *Tryanuka* is composed of exactly six primary *Paramanus*. Higher macroscopic bodies are subsequently constructed by the aggregation of numerous *Tryanukas*.

Chemical Transformation under Heat (*Paka*)

The Vaisheshika system explained chemical changes—such as the ripening of fruit, the baking of clay pots, or the oxidation of metals—through thermal action (*Paka*). Two distinct theories arose to explain the microscopic mechanism of heat-induced transformation:

1. **Pilu-paka (Atomic Transformation)**: Championed by the Vaisheshika school, this model posits that when a body (such as a raw clay pot) is subjected to intense fire (*Tejas*), the heat breaks the bonds between the constituent *Dyanukas* and *Tryanukas*, disintegrating the body completely into free *Paramanus*. The thermal energy acts directly on these individual *Paramanus*, altering their color, texture, and density. Upon cooling, the transformed *Paramanus* recombine into new dyads and triads, yielding a chemically transformed macroscopic object.
2. **Pithara-paka (Molecular Transformation)**: Championed by the *Nyaya* school, this alternative view posited that the structural framework of the body remains intact during heating. Thermal energy penetrates the pores of the macroscopic body (*Pithara*) as a whole, altering the internal properties of its constituent molecules simultaneously without disassembling the body into isolated atoms.

Chemical Knowledge Preserved in Ancient Indian Texts

Systematic chemical knowledge was documented and preserved in foundational Sanskrit treatises dealing with statecraft, metallurgy, and medicine.

Kautilya's Arthashastra

Written around the 4th century BCE by Kautilya (Chanakya), the *Arthashastra* provides detailed documentation of administrative, industrial, and technological state operations in ancient India.

A significant portion of the text is devoted to the *Akaradhyaksha* (Superintendent of Mines) and the *Lohadhyaksha* (Superintendent of Metals). Key chemical and metallurgical aspects recorded include: * **Identification of Ores**: The text describes physical characteristics of metallic ores (*Dhatu*) found in riverbeds, caves, and underground seams—noting their

color, weight, greasy feel, and smell. Gold-bearing ores are described as yellowish or rose-colored, while copper ores are noted for their heavy, greenish-blue appearance. * **Extraction and Refining**: It details methods of washing ores to remove earthy impurities (*gangue*), roasting sulfide ores to expel sulfur dioxide, and smelting using blast furnaces powered by leather bellows with charcoal as the reducing agent. * **Assaying and Touchstone Tests**: To determine the purity of gold and silver, touchstones (*Nikasha*) were used. The color of the streak left on the touchstone when treated with specific chemical reagents or heat provided a quantitative measure of purity. * **Alloys and Minting**: Processes for alloying copper, tin, lead, and zinc to produce bronze, brass, and coinage alloys of precise composition are explicitly outlined.

The Charaka Samhita and Sushruta Samhita

The foundational medical treatises of Ayurveda—the *Charaka Samhita* (focused on internal medicine) and the *Sushruta Samhita* (focused on surgery), compiled between the 6th century BCE and the 2nd century CE—contain extensive sections on pharmaceutical chemistry.

1. **Charaka Samhita**:

* Describes the processing of plant-based drugs, herbal decoctions, and tinctures. * Details the preparation of medicinal wines through controlled fermentation (*Asavas* and *Arishtas*). * Outlines the early use of metallic powders (*Bhasmas*) produced by calcining metals (iron, copper, gold) with plant juices and sulfur, rendering them non-toxic and absorbable by the human body.

1. **Sushruta Samhita**:

* Presents a systematic classification of alkalis (*Kshara*) into three operational strengths: *Mridu* (mild), *Madhyama* (medium), and *Tikshna* (highly caustic/corrosive). * Details the precise laboratory method for preparing caustic alkalis: plant ash is dissolved in water, filtered through cloth twenty-one times, mixed with burnt limestone (CaO) or calcined oyster shells, and boiled down in iron vessels to form concentrated caustic hydroxides (KOH or NaOH). The chemical reactions involved are represented by the slaking of quicklime in water:

$$\mathrm{CaO} + \mathrm{H_2O} \longrightarrow \mathrm{Ca(OH)_2}$$

This is followed by the double displacement caustification reaction with potassium carbonate present in plant ash:

$$\mathrm{K_2CO_3} + \mathrm{Ca(OH)_2} \longrightarrow 2\,\mathrm{KOH} + \mathrm{CaCO_3}\downarrow$$

The insoluble calcium carbonate ($\mathrm{CaCO_3}$) precipitates out, leaving a concentrated solution of caustic potassium hydroxide (KOH). * Explains surgical application of these caustics for cauterizing wounds, sterilizing instruments, and treating lesions.

Continuity of Chemical Traditions into Later Periods

The chemical foundations established during the Vedic and classical eras did not remain static; they maintained an unbroken continuity, expanding significantly into later historical periods.

By the early medieval period (c. 5th–12th century CE), empirical chemical operations evolved into the formal discipline of *Rasashastra* (alchemical chemistry). Textual treatises such as the *Rasaratnakara* of Nagarjuna, *Rasarnava*, and *Rasaratnasamuccaya* expanded upon ancient laboratory procedures. They introduced standardized chemical apparatuses (*Yantras*), heating crucibles (*Musas*), specialized furnaces (*Kosthis*), and advanced techniques for the distillation, sublimation, and cupellation of mercury, zinc, sulfur, and metallic oxides.

Simultaneously, traditional chemical crafts—such as high-grade iron and Wootz steel metallurgy, glass manufacture, ceramic glazing, perfumery (*Attar* distillation), and natural textile dyeing—flourished in artisanal guilds across the Indian

subcontinent. The continuous transmission of practical laboratory methods alongside theoretical atomic models ensured that ancient Indian chemical traditions served as a vital precursor to early modern chemical sciences.

2. Ayurvedic Chemistry, Alchemy and Chemistry in Medical Schools of Ancient India

Ayurvedic Chemistry

Ayurvedic chemistry represents one of the earliest systematic applications of chemical operations to therapeutic sciences. The philosophical and practical framework of Ayurveda rests upon the balance of three fundamental physiological humours or *doshas*: *Vata* (governing biological movement and transport), *Pitta* (governing metabolic transformations and thermal regulation), and *Kapha* (governing structural stability and fluid balance). Pathological conditions are understood as imbalances among these *doshas*. To restore physiological equilibrium, ancient practitioners utilized natural substances categorized into plant, mineral, and animal products.

Plant-derived materials included leaves, roots, barks, seeds, and exudates containing complex secondary metabolites such as alkaloids, flavonoids, and essential oils. Animal-derived preparations incorporated substances like milk, ghee, honey, animal bile, and calcified shells (such as cowrie and pearl) rich in calcium carbonate. Mineral substances encompassed elemental metals, ores, metallic salts, and sulfur compounds.

To transform raw materials into bioavailable forms, classical Ayurveda established distinct pharmaceutical techniques for extraction and formulation:

1. **Decoctions (*Kwatha*):** Coarsely powdered plant material is boiled with a specified volume of water—typically sixteen parts of water to one part of herb—and reduced over controlled heat until one-fourth of the original fluid volume remains. This thermal extraction dissolves water-soluble active principles while concentrating non-volatile phytochemicals.
2. **Powders (*Churna*):** Cleaned, dried botanical or mineral constituents are finely ground and sieved to uniform particle size. Micro-particulate size reduction increases total surface area, facilitating rapid enzymatic hydrolysis and absorption in the gastrointestinal tract.
3. **Pastes (*Kalka*):** Fresh or dried plant parts are ground with a minimal volume of liquid (water, juice, or oil) to form a homogeneous viscous mass, extracting both polar and non-polar constituents into a moist matrix.
4. **Fermented Preparations (*Asava* and *Arishta*):** *Arishta* formulations are prepared by fermenting an aqueous decoction (*kwatha*), whereas *Asava* formulations utilize fresh unboiled plant juices or herbal infusions. In both methods, natural sugars (such as jaggery, honey, or dried grapes) and fermenting flowers (*Woodfordia fruticosa*) are added to sealed wooden or earthenware vessels. Anaerobic fermentation by native yeasts converts simple carbohydrates into ethanol. The self-generated alcohol (typically $5–12\%$ by volume) acts as a natural solvent that extracts fat-soluble and water-soluble active compounds while functioning as a preservative.

In *Rasa Shastra* (the branch of Ayurveda devoted to mineral processing), elemental mercury (*Rasa* or *Parada*) and sulfur (*Gandhaka*) hold central chemical importance. Metallic mercury is liquid at ambient conditions and forms amalgams with numerous metals. To eliminate inherent toxicities (*doshas* of raw mercury), mercury undergoes rigorous purification processes (*Shodhana*). The fundamental chemical reaction uniting mercury and sulfur involves the trituration of purified mercury with elemental sulfur in equal or specified stoichiometric ratios without heat, producing a fine black powder known as *Kajjali* (black mercuric sulfide).

The solid-state reaction between elemental mercury and sulfur yields mercuric sulfide:

$$\mathrm{Hg\ (l) + S\ (s) \rightarrow HgS\ (s)}$$

This black mercuric sulfide (β-HgS, metacinnabar) serves as the non-toxic chemical precursor for further thermal processing, such as sublimation into red cinnabar (α-HgS or *Rasa-sindura*), which acts as a potent therapeutic agent.

Calcination (*Bhasma*) of Metals and Minerals

A key achievement of *Rasa Shastra* is *Bhasmikarana*, a specialized calcination procedure that converts dense, toxic metals into bio-compatible, ultra-fine nano-particulate inorganic oxides or sulfides termed *Bhasmas*. The process involves two primary steps: *Shodhana* (detoxification/purification through thermal quenching in organic liquids like milk, buttermilk, or herbal juices) and *Marana* (incineration through repeated closed heating cycles known as *Puta*).

In the preparation of Gold Bhasma (*Swarna Bhasma*), high-purity metallic gold is leafed into extremely thin foils, heated to red heat, and quenched in various liquid media to induce structural stress and surface modification. The purified gold leaves are triturated with purified mercury to form a gold-mercury amalgam. Pure sulfur or herbal juices (such as *Aloe vera*) are added, and the triturated paste is sealed in shallow earthen crucibles (*Sharava Samputa*) with a clay-cloth lute. The sealed crucible is subjected to high thermal energy in a pit furnace. The mercury volatilizes and escapes or acts as a reaction catalyst, while gold undergoes grain refinement down to the nanoscale level ($20–50\ \mathrm{nm}$), yielding stabilized metallic gold nanoparticles coated with trace organic ligands.

In the preparation of Iron Bhasma (*Lauha Bhasma*), elemental iron filings are purified by heating and quenching in decoctions of *Triphala*. The purified iron is triturated with sulfur and plant juices (such as *Lemon juice* or *Eclipta alba*) to yield an intermediate sulfide-hydroxide paste. The mixture is sealed in earthen crucibles and subjected to calcination cycles at temperatures ranging from $600^\circ\mathrm{C}$ to $800^\circ\mathrm{C}$.

During early thermal exposure in the presence of atmospheric oxygen within the crucible, elemental iron reacts with oxygen to form ferrous oxide:

$$2\mathrm{Fe\ (s)} + \mathrm{O_2\ (g)} \rightarrow 2\mathrm{FeO\ (s)}$$

Upon continued high-temperature heating and repeated oxidation cycles (*Puta*), ferrous oxide undergoes further oxidation to produce magnetic iron oxide (magnetite):

$$6\mathrm{FeO\ (s)} + \mathrm{O_2\ (g)} \rightarrow 2\mathrm{Fe_3O_4\ (s)}$$

With prolonged calcination, magnetite fully oxidizes into stable ferric oxide (hematite):

$$4\mathrm{Fe_3O_4\ (s)} + \mathrm{O_2\ (g)} \rightarrow 6\mathrm{Fe_2O_3\ (s)}$$

The final *Lauha Bhasma* consists predominantly of micro- and nano-crystalline ferric oxide (α-$\mathrm{Fe_2O_3}$), which is non-magnetic, red-brown, free of free metallic iron, and readily soluble in gastric acid.

Bhasmas are characterized by their unique properties: *Varitaratvam* (ability to float on water due to low density and high surface area), *Rekhapurna* (particles fine enough to enter the crevices of human fingerprints), and *Apunarbhava* (inability to be reduced back to the parent metal upon heating with carbonaceous reducing agents). Medicinally, *Swarna Bhasma* is utilized as a rejuvenator, anti-inflammatory, and neuroprotective agent, while *Lauha Bhasma* is employed to treat iron-deficiency anemia (*Pandu*) and metabolic disorders.

Unit Operations in Ayurvedic Drug Preparation

Ancient Indian pharmaceutical chemistry relied on three fundamental unit operations to isolate and purify active substances:

1. **Extraction:** Solute extraction was carried out using polar liquids (water, milk, vinegars) and non-polar media (clarified butter, oils). Lipid-based extractions (*Snehapaka*) involved heating plant materials in sesame oil or ghee alongside aqueous decoctions. Water-soluble compounds were extracted into the aqueous layer, while lipophilic principles dissolved

into the lipid phase. Continued heating evaporated the aqueous component, yielding a stabilized lipid formulation enriched with fat-soluble phytochemicals.

2. **Distillation (*Tiryak-patana* and *Urdhva-patana*):** Distillation apparatuses, such as the *Tiryak-patan yantra* (slanted/lateral distillation unit) and *Vidyadhara yantra* (vertical condensation unit), were constructed using earthenware vessels fitted with water-cooled receivers. Liquid mixtures or herbal suspensions were heated in a lower pot, forcing volatile vapors upward or laterally through delivery tubes. Condensation occurred on cooler surfaces above or adjacent to the heat source. This method was widely used to extract aromatic essential oils, volatile plant acids, and pure liquid mercury from cinnabar ore (HgS).
3. **Sublimation (*Urdhva-patana*):** Sublimation was employed to purify volatile solids such as camphor (*Karpura*), ammonium chloride (*Navasadara*), and elemental mercury. The crude material was placed in the bottom vessel of a sealed two-tier earthen apparatus. Heating the lower vessel caused the solid to transition directly into the gas phase, bypassing the liquid state. The gas condensed as a purified crystalline deposit on the underside of the upper, water-cooled vessel surface.

Alchemy in India (Rasashastra)

Foundations of Rasashastra and the Alchemical Tradition

Rasashastra emerged as an integrated system of alchemical philosophy, chemical technology, and pharmacology. The discipline pursued two main goals: *Deha-siddhi* (the transformation and preservation of the human body through age-defying elixirs, known as *Rasayana*) and *Loha-siddhi* (the transmutation of base metals such as lead, copper, and iron into noble metals like silver and gold, known as *Loha-vedha*).

The theoretical foundation of *Rasashastra* rested on the belief that mercury (*Rasa*), considered the divine fluid of Lord Shiva, possessed unique transformative capabilities when stabilized or "killed" (*Murchhana* or *Marana*) through complex chemical treatments involving sulfur (*Gandhaka*), mica (*Abhraka*), and plant extracts. Transmutation served both as an economic pursuit and as a rigorous experimental benchmark: if an alchemical process could successfully alter the physical properties and atomic structure of base metals to match gold, the resulting mercurial preparation was deemed sufficiently potent to transform human physiology and grant longevity.

Contributions of Nagarjuna and the *Rasaratnakara*

The premier historical figure in Indian alchemy is Nagarjuna (c. 8th–10th century CE), a master chemist whose foundational text, the *Rasaratnakara*, documented early experimental chemistry. The *Rasaratnakara* contains systematic descriptions of chemical operations, laboratory apparatus (*Yantras*), crucibles (*Musha*), furnaces (*Koshthi*), and chemical reagents (*Vidas*).

Nagarjuna detailed the preparation of mercury compounds, the extraction of metals from natural ores, and the use of vegetable solvents and mineral acids. He introduced specialized apparatus such as the *Dolayantra* (suspension apparatus for gentle heating and digestion), the *Svedani yantra* (steaming apparatus), and the *Patana yantra* (distillation apparatus). Nagarjuna's work established standard chemical terminology and documented early recipes for zinc distillation, brass synthesis, and the production of colored metal alloys designed to mimic gold and silver.

The Eight Metals (*Ashta Dhatu*): Properties, Classification, and Purification

Rasashastra categorizes metals into primary groups, recognizing eight principal metals (*Ashta Dhatu*):

1. *Swarna* (Gold)
2. *Rupya* (Silver)
3. *Tamra* (Copper)

4. *Lauha* (Iron)
5. *Vanga* (Tin)
6. *Naga* (Lead)
7. *Yashada* (Zinc)
8. *Rasa* (Mercury)

These metals were classified based on their density, malleability, melting point, reactivity, and resistance to corrosion. Gold and silver were recognized as noble metals (*Suddha Loha*) due to their chemical inertness and resistance to oxidation at high temperatures. Copper, iron, tin, lead, and zinc were categorized as base metals (*Puti Loha* or *Kshaya Loha*) because they oxidize, tarnish, or volatilize upon heating. Mercury was treated as the sovereign element (*Rasa-raj*) due to its unique liquid state.

Because crude metals contain toxic impurities, *Rasashastra* required mandatory purification (*Shodhana*) prior to any therapeutic application. *Shodhana* combines physical surface cleaning with chemical surface modification. The standard process involves heating the metal to a molten or red-hot state and quenching (*Nirvapana*) it into specified liquid media, such as plant juices, buttermilk, sesame oil, cow's urine, or herbal decoctions.

During thermal quenching, the sudden drop in temperature generates intense mechanical thermal shock:

$$\Delta T = T_{\text{hot}} - T_{\text{liquid}}$$

This rapid contraction causes internal strain and fragmentation of the solid metal matrix, cracking the bulk material into fine micro-particles. Simultaneously, organic acids, tannins, and sulfur-containing compounds present in the organic quenching media react with the hot metal surface, forming soluble metal salts or surface complexes that dissolve trace toxic impurities (such as arsenic, cadmium, or antimony). Repeated quenching yields a clean, brittle, finely divided metallic precursor ready for calcination.

Preparation and Application of Acids, Alkalis, and Mineral Salts

Ancient Indian alchemists synthesized mineral acids (*Dravaka*), strong alkalis (*Kshara*), and mineral salts for dissolving metals, preparing mineral reagents, and carrying out transmutation reactions.

Mineral acids were produced by dry distillation of mineral salts. Nitric acid (*Sankha Dravaka*) was prepared by mixing potassium nitrate (*Suryakshara*), ferrous sulfate (*Kasisa*), and alum (*Sphatika*) in an earthenware distillation vessel and applying intense heat.

The thermal decomposition of ferrous sulfate produces iron(III) oxide, sulfur dioxide, and sulfur trioxide:

$$2\text{FeSO}_4\ (\text{s}) \xrightarrow{\Delta} \text{Fe}_2\text{O}_3\ (\text{s}) + \text{SO}_2\ (\text{g}) + \text{SO}_3\ (\text{g})$$

The liberated sulfur trioxide gas reacts with water vapor present in the clay matrix or receiver to form sulfuric acid:

$$\text{SO}_3\ (\text{g}) + \text{H}_2\text{O}\ (\text{g}) \rightarrow \text{H}_2\text{SO}_4\ (\text{l})$$

Potassium nitrate reacts with the generated sulfuric acid to produce potassium hydrogen sulfate and nitric acid vapor:

$$\text{KNO}_3\ (\text{s}) + \text{H}_2\text{SO}_4\ (\text{l}) \xrightarrow{\Delta} \text{KHSO}_4\ (\text{s}) + \text{HNO}_3\ (\text{g})$$

The gaseous nitric acid (HNO_3) condenses in the receiver vessel as a concentrated acidic distillate capable of dissolving metals such as silver and copper. Aqua regia (*Kasa-drava*), capable of dissolving noble gold, was formulated by distilling potassium nitrate and ammonium chloride (*Navasadara*) with green vitriol.

Strong alkalis (*Kshara*) were prepared from plant ashes rich in potassium carbonate ($\mathrm{K_2CO_3}$) and sodium carbonate ($\mathrm{Na_2CO_3}$), such as *Yavakshara* (obtained from barley ash) and *Svarjikakshara* (obtained from natron or halophytic plant ash). Caustic alkalis (such as caustic soda, NaOH, or caustic potash, KOH) were synthesized through causticization with quicklime (CaO).

Calcite or limestone ($\mathrm{CaCO_3}$) is thermally decomposed to yield quicklime:

$$\mathrm{CaCO_3\ (s) \xrightarrow{\Delta} CaO\ (s) + CO_2\ (g)}$$

The resulting quicklime is slaked with water to form calcium hydroxide:

$$\mathrm{CaO\ (s) + H_2O\ (l) \rightarrow Ca(OH)_2\ (aq)}$$

The aqueous calcium hydroxide is mixed with a solution of sodium carbonate:

$$\mathrm{Ca(OH)_2\ (aq) + Na_2CO_3\ (aq) \rightarrow 2NaOH\ (aq) + CaCO_3\ (s)}$$

Precipitated calcium carbonate ($\mathrm{CaCO_3}$) is removed by filtration, leaving a strongly alkaline solution of caustic soda (NaOH). These alkalis were classified into mild (*Mridu*), medium (*Madhyama*), and extremely caustic (*Tikshna*) grades based on the amount of quicklime added, and were used for surgical cauterization, dissolution of tissues, and cleansing of metal surfaces.

Chemistry in Medical Schools of Ancient India

Academic Institutions: Taxila and Nalanda

Ancient Indian medical education flourished at major university centers such as Taxila (Takshashila, located in modern Punjab) and Nalanda (located in modern Bihar). These institutions contained specialized departments, extensive libraries, botanical gardens, and dedicated chemical laboratories (*Rasa-shalas*).

The *Rasa-shalas* were constructed according to precise technical guidelines. Texts like the *Rasaratnasamuccaya* specify that the laboratory should be oriented toward the east or north-east, equipped with clean water sources, and arranged into functional zones: - The east sector was reserved for furnace operations (*Koshthi*), - The south-west for grinding and trituration (*Mardan*), - The north-west for distillation and sublimation apparatus, - The central zone for raw material storage and analytical inspection.

At Taxila and Nalanda, students underwent multi-year instruction combining theoretical study with hands-on laboratory practice. Medical scholars learned to identify mineral ores, extract metal values, prepare chemical reagents, calibrate thermal processing pits (*Putas*), and standardize drug formulations.

Foundational Treatises: *Charaka Samhita* and *Sushruta Samhita*

The primary theoretical foundations of ancient Indian medicine and surgery are recorded in the *Charaka Samhita* (attributed to Charaka, focusing on internal medicine) and the *Sushruta Samhita* (attributed to Sushruta, focusing on surgical techniques and metallurgy).

The *Charaka Samhita* presents a systematic chemical classification of therapeutic agents based on their origin (botanical, animal, or mineral), taste (*Rasa*), active potency (*Virya*), post-digestive metabolic transformation (*Vipaka*), and specific pharmacological action (*Prabhava*). Drugs were formulated using precise stoichiometric ratios of vehicle substances (*Anupana*), such as milk, honey, warm water, or sesame oil, which served as lipidic or aqueous carriers to optimize drug transport across biological membranes.

The *Sushruta Samhita* focuses on surgical procedures (*Sastrakarma*), anatomical dissection, and wound management. Sushruta classified surgical tools into two main categories: *Yantras* (blunt instruments, forceps, clamps) and *Sastras*

(sharp instruments, scalpels, lancets, needles). The text describes over 120 surgical instruments, detailing their dimensions, functional ergonomics, and specific metallurgical fabrication techniques.

Metallurgy of Surgical Instruments, Anaesthetics, Antiseptics, and Dressings

The manufacture of surgical instruments described by Sushruta required advanced iron and steel metallurgy. Scalpels, lancets, and needles were forged from high-carbon steel (*Wootz steel* or *Karavira-patra* iron). The iron was melted with carbonaceous plant materials (such as wood of *Cassia auriculata* and leaves of *Calotropis*) in closed earthen crucibles, yielding a uniform high-carbon steel alloy with a high carbon content ($1.0–1.5\ \text{wt}\%\ \text{C}$).

To achieve a sharp, durable cutting edge, instruments underwent a specialized heat treatment process known as *Pana* (quenching and tempering). The forged steel blade was heated to an elevated austenitizing temperature until it glowed bright cherry-red:

$$\text{Steel (ferrite + cementite)} \xrightarrow{\Delta} \text{Steel (austenite, } \gamma\text{-Fe)}$$

The red-hot blade was quenched in specific liquid baths, such as cold water, saline solutions, alkali extracts, or plant oils. Rapid quenching converts the face-centered cubic austenite (γ-Fe) into a body-centered tetragonally distorted, hard, brittle phase known as martensite:

$$\gamma\text{-Fe (austenite)} \xrightarrow{\text{rapid quenching}} \text{Martensite}$$

Following quenching, the blade was tempered by mild reheating to relieve internal stress and reduce brittleness, balancing hardness with fracture toughness. Instruments were tested for sharpness by cutting fine hair filaments or slicing through floating lotus stems.

For pain management during surgery, ancient Indian physicians employed inhaled and ingested phytopharmaceutical preparations as anaesthetics and sedatives. Formulations containing *Sammohani* (alcohol-enriched botanical infusions), high doses of *Cannabis sativa* (*Bhanga*), *Papaver somniferum* (opium, *Ahiphena*), and *Datura stramonium* (containing anticholinergic tropane alkaloids like scopolamine and atropine) were administered to induce central nervous system depression, muscle relaxation, and analgesia prior to major surgical procedures.

Antiseptics and wound dressings were formulated using natural antimicrobial agents. Wounds (*Vrana*) were cleansed with warm aqueous decoctions of tannin-rich plants, such as *Neem* (*Azadirachta indica*), *Panchavalkala* (barks of five *Ficus* species), and *Turmeric* (*Curcuma longa*). Turmeric contains the polyphenolic diketone compound curcumin, which exhibits antimicrobial and anti-inflammatory activity.

Dressings (*Vrana-bandhana*) incorporated lint or cotton cloths impregnated with medicated fats (*Ghrita* or *Taila*) containing bees' wax, resin from *Shorea robusta* (*Rala*), zinc salts (*Tuttha*), and copper compounds (*Kasisa*). The applied lipids formed a protective hydrophobic barrier, preventing bacterial contamination and maintaining a moist environment that accelerated tissue regeneration.

Role of Chemistry in Diagnosis and Treatment

Chemistry played a central diagnostic role in ancient Indian clinical practice. Diagnostic protocols included the chemical analysis of metabolic waste products, particularly urine (*Mutra-pariksha*). Physicians analyzed physical and chemical properties of urine, including color, clarity, odor, specific gravity, sediment formation, and taste.

The diagnosis of *Madhumeha* (literally "honey-urine," corresponding to Diabetes Mellitus) relied on observable chemical and biological tests. Physicians noted that urine from affected patients possessed a sweet taste, attracted ants and insects, and exhibited elevated viscosity and density:

$$\text{Elevated glucose concentration} \rightarrow \text{Increased urine specific gravity and sweetness}$$

This early biochemical observation directly correlated glycosuria with metabolic dysfunction. Similarly, variations in urine color and turbidity were linked to specific metabolic disturbances: dark yellow-brown urine containing bile pigments indicated hepatic involvement (*Kamala* or jaundice), while red-tinged urine indicated hematuria.

Therapeutically, ancient physicians applied chemical principles through customized drug delivery systems. The choice of therapeutic agents was guided by neutralizing metabolic imbalances through acid-base and mineral chemistry. Hyperacidity (*Amlapitta*) was treated using neutralizing mineral bases such as *Sankha Bhasma* (calcined conch shell, consisting of calcium oxide and calcium carbonate, $\mathrm{CaCO_3}$) and *Pravala Bhasma* (calcined coral).

The basic calcium salts neutralize excess hydrochloric acid (HCl) in the gastric mucosa:

$$\mathrm{CaCO_3\ (s) + 2HCl\ (aq) \rightarrow CaCl_2\ (aq) + H_2O\ (l) + CO_2\ (g)}$$

This acid-base neutralization reaction relieved gastric irritation and restored physiological pH balances.

Through these systematic methods of drug preparation, alchemical transformations, and clinical applications, ancient Indian medical schools established an empirical foundation that integrated fundamental chemistry with therapeutic medicine.

3. Metal and Metallurgy in Ancient India

The development of metallurgy in ancient India represents one of the most remarkable chapters in the history of chemical science and engineering. Indian metallurgists pioneered advanced techniques for mining, smelting, refining, and alloy preparation long before these methods were codified in Western industrial traditions. The ability to extract pure metals from complex mineral ores required a sophisticated understanding of pyrotechnology, reduction chemistry, and temperature control within specialized furnaces.

Mining and Extraction of Metals

Mining operations in ancient India were conducted with notable geological foresight. Underground shafts and galleries were constructed to reach deep ore deposits, particularly in regions rich in metalliferous minerals such as Rajasthan, Bihar, and South India. The extraction process involved breaking ore-bearing rocks using fire-setting techniques followed by thermal quenching with water, causing structural fracturing due to sudden thermal contraction.

The crushed ores were separated from gangue minerals by gravity separation, roasting, and washing. The chemical reduction of metallic oxides to free metals was carried out in custom-built clay furnaces using wood charcoal as both the heat source and the reducing agent.

Iron Metallurgy and Wootz Steel

Ancient Indian iron masters developed two distinct traditions of iron production: bloomery iron (wrought iron) and ultra-high carbon crucible steel, world-renowned as *wootz steel*.

In the bloomery furnace process, hematite ($\mathrm{Fe_2O_3}$) or magnetite ($\mathrm{Fe_3O_4}$) ore was layered with charcoal inside a vertical shaft furnace made of clay. Air was supplied through hand-operated tuyères using leather bellows. Carbon monoxide produced by the incomplete combustion of charcoal reduced the iron ore solid-state without fully melting the iron:

$$\mathrm{C(s) + O_2(g) \rightarrow CO_2(g)}$$

$$\mathrm{CO_2(g) + C(s) \rightarrow 2CO(g)}$$

The carbon monoxide gas reacted with ferric oxide in successive reduction steps:

$$\mathrm{3Fe_2O_3(s) + CO(g) \rightarrow 2Fe_3O_4(s) + CO_2(g)}$$

$$\mathrm{Fe_3O_4(s) + CO(g) \rightarrow 3FeO(s) + CO_2(g)}$$

$$\mathrm{FeO(s) + CO(g) \rightarrow Fe(s) + CO_2(g)}$$

The solid metallic iron coalesced into a porous mass termed a 'bloom', while siliceous gangue reacted with ferrous oxide to form a fluid liquid slag:

$$\mathrm{FeO(s) + SiO_2(s) \rightarrow FeSiO_3(l)}$$

The solid iron bloom was hammered repeatedly at high temperatures to squeeze out the trapped liquid slag, producing malleable wrought iron.

A major achievement of ancient Indian iron metallurgy is the **Delhi Iron Pillar**, forged around the 4th century CE during the Gupta period. Standing over 7 meters tall and weighing more than 6 tonnes, it demonstrates extraordinary resistance to atmospheric corrosion. Microstructural and chemical analysis reveals that the pillar was manufactured by forge-welding blooms of wrought iron.

The rust resistance of the Delhi Iron Pillar is attributed to its high phosphorus content (up to 0.25% by weight) and low sulfur and manganese levels. The elevated phosphorus content promotes the formation of a protective passive film on the metal surface. In atmospheric conditions, iron undergoes electrochemical oxidation to form an initial rust layer:

$$\mathrm{2Fe(s) + O_2(g) + 2H_2O(l) \rightarrow 2Fe(OH)_2(s)}$$

Further oxidation yields hydrated ferric oxyhydroxide:

$$\mathrm{4Fe(OH)_2(s) + O_2(g) \rightarrow 4FeOOH(s) + 2H_2O(l)}$$

In the presence of dissolved phosphate ions catalyzed by the high phosphorus concentration of the iron matrix, the amorphous oxyhydroxide layer transforms over time into a dense, non-porous, protective crystalline barrier layer of iron hydrogen phosphate hydrate ($\mathrm{FePO_4 \cdot H_3PO_4 \cdot 4H_2O}$) and misawite ($\delta$-$\mathrm{FeOOH}$). This passive layer acts as an impermeable barrier against oxygen and moisture diffusion, arresting further corrosion.

Wootz steel represents a high-carbon alloy ($1.0 - 1.6\%$ carbon by weight) manufactured in South India (Golconda, Tamil Nadu, and Karnataka) since the 1st millennium BCE. Carbon-free bloomery iron was cut into small pieces and sealed inside airtight clay crucibles along with carbonaceous plant materials, such as the leaves of *Calotropis gigantea* and wood of *Cassia auriculata*. The sealed crucibles were heated in specialized furnaces to temperatures exceeding $1200^\circ\mathrm{C}$ for several hours.

During this process, carbon absorbed into the solid iron, lowering its melting point according to the iron-carbon phase diagram. The carbon dissolved uniformly into the iron phase:

$$\mathrm{Fe(s) + C(organic) \rightarrow Fe\text{-}C\ (austenite\ solution)}$$

Upon slow controlled cooling inside the furnace, microscopic crystals of cementite ($\mathrm{Fe_3C}$) precipitated out within a pearlitic matrix. The forged wootz steel exhibited characteristic swirling patterns of carbide particles, providing an ideal combination of hardness, toughness, and sharp edge retention. Wootz steel ingots were traded globally for making legendary Damascus sword blades.

Copper Metallurgy and the Copper Hoard Culture

Copper was one of the earliest metals extracted in India, with extensive mining evidence found in the Khetri copper belt of Rajasthan and the Singhbhum region of Jharkhand. The Chalcolithic period and the widespread *Copper Hoard culture* (circa 2000–1500 BCE) produced large quantities of copper implements, including celts, harpoons, swords, and anthropomorphic figures.

Copper extraction relied on processing copper carbonate ores like malachite ($\mathrm{Cu_2CO_3(OH)_2}$) or sulfide ores like chalcopyrite ($\mathrm{CuFeS_2}$). For carbonate ores, thermal decomposition yielded copper(II) oxide:

$$\mathrm{Cu_2CO_3(OH)_2(s) \rightarrow 2CuO(s) + CO_2(g) + H_2O(g)}$$

The oxide was reduced by reaction with charcoal inside small clay shaft furnaces:

$$\mathrm{CuO(s) + C(s) \rightarrow Cu(l) + CO(g)}$$

For sulfide ores, the ore was first roasted in air to convert sulfides into oxides and sulfates:

$$\mathrm{2CuFeS_2(s) + 7O_2(g) \rightarrow 2CuO(s) + 2FeO(s) + 4SO_2(g)}$$

Smelting with silica flux ($\mathrm{SiO_2}$) removed the iron oxide as silicate slag:

$$\mathrm{FeO(s) + SiO_2(s) \rightarrow FeSiO_3(l)}$$

The residual copper oxide was subsequently reduced to metallic copper, which settled at the bottom of the furnace as a heavy liquid layer.

Pyrotechnological Distillation of Zinc at Zawar

The extraction of metallic zinc is historically one of the most challenging metallurgical processes because the reduction temperature of zinc oxide ($\sim 950^\circ\mathrm{C}$) is higher than the boiling point of elemental zinc metal ($907^\circ\mathrm{C}$). Consequently, zinc is produced in gaseous form and rapidly re-oxidizes if exposed to carbon dioxide or oxygen during cooling.

Ancient Indian metallurgists at **Zawar in Rajasthan** solved this problem in the medieval period (from about the 12th century CE), establishing the world's first industrial-scale pyrotechnology for downward distillation of zinc.

The zinc ore, primarily sphalerite (ZnS) or calamine ($\mathrm{ZnCO_3}$), was roasted to yield zinc oxide:

$$\mathrm{ZnCO_3(s) \rightarrow ZnO(s) + CO_2(g)}$$

$$\mathrm{2ZnS(s) + 3O_2(g) \rightarrow 2ZnO(s) + 2SO_2(g)}$$

The roasted zinc oxide was mixed with organic reductants (charcoal, salts, and cow dung) and loaded into specialized eggplant-shaped clay retorts (*brinjal-shaped retorts*). A long clay condenser tube was attached to the open mouth of each retort.

The retorts were placed upside down in a custom two-tier furnace called a *Tiryak-patana-yantra*. The upper chamber containing the retorts was heated intensely by a charcoal fire to temperatures around $1100^\circ\mathrm{C}$. The carbothermic reduction proceeded internally:

$$\mathrm{ZnO(s) + C(s) \rightarrow Zn(g) + CO(g)}$$

The zinc vapor generated inside the retort passed downward through the ceramic condenser tube into a lower, cooler collection chamber. As the zinc vapor cooled below $907^\circ\mathrm{C}$ in a reducing carbon monoxide atmosphere, it condensed directly into liquid zinc droplets, avoiding re-oxidation:

$$\mathrm{Zn(g) \xrightarrow{cooling} Zn(l)}$$

The liquid zinc collected into earthen vessels placed beneath the condenser tubes. This downward distillation process produced pure metallic zinc centuries before similar techniques were patented in Europe in the 18th century.

Gold and Silver Metallurgy

Gold (Au) and silver (Ag) were refined using physical separation and chemical cupellation. Native alluvial gold was collected from riverbeds through panning and gravity washing. Deep vein quartz mining for gold was also practiced at Kolar and Hutti in Karnataka.

Silver was primarily extracted from argentiferous galena (PbS carrying silver impurities). The sulfide ore was roasted and smelted to yield a lead-silver alloy. Purification of silver was achieved via **cupellation**. The alloy was melted in a porous bone-ash or clay crucible under an oxidizing atmosphere. Lead oxidized rapidly to lead monoxide (litharge):

$$\mathrm{2Pb(l) + O_2(g) \rightarrow 2PbO(l)}$$

The molten lead oxide (PbO) was absorbed into the porous walls of the cupel or scraped away, leaving behind a button of pure metallic silver.

Smelting Furnaces and Techniques

Ancient Indian metallurgy relied on a variety of furnace designs engineered to achieve high temperatures and specific chemical atmospheres:

1. **Shaft Furnaces:** Vertical cylindrical clay furnaces powered by natural draft or forced air bellows. Used for bloomery iron and copper extraction.
2. **Crucible Furnaces:** Specialized closed furnaces capable of maintaining uniform temperatures above $1200^\circ\mathrm{C}$ for extended periods, essential for wootz steel production.
3. **Downward Distillation Furnaces (*Tiryak-patana-yantra*):** Perforated hearth furnaces designed with separate heating and condensation zones specifically engineered for volatile metals like zinc.

Alloys of Ancient India

By controlling the stoichiometry of base metals and fluxing agents, ancient Indian metallurgists produced alloys with precise mechanical, anti-corrosive, and aesthetic properties.

Bronze (Copper + Tin)

Bronze was produced by melting copper together with tin. The addition of tin lowers the melting point of copper from $1085^\circ\mathrm{C}$ to below $1000^\circ\mathrm{C}$ while increasing hardness and castability:

$$\mathrm{Cu(l) + } x\,\mathrm{Sn(l) \rightarrow Cu\text{-}Sn\ alloy\ (Bronze)}$$

High-tin bronzes ($20 - 25\%$ tin) were crafted into thin-walled ringing bowls and mirrors by hot-forging followed by rapid quenching to stabilize the ductile β-phase alloy structure. The lost-wax technique (*cire perdue*) was extensively used to cast detailed bronze icons, such as the Chola bronze sculptures.

Brass (Copper + Zinc)

Brass was initially prepared by *cementation*, wherein copper sheets were heated in contact with roasted zinc ore (ZnO) and charcoal in a sealed crucible. The zinc vapor absorbed directly into the solid copper phase:

$$\mathrm{ZnO(s) + C(s) + Cu(s) \rightarrow Cu\text{-}Zn\ alloy\ (Brass) + CO(g)}$$

After the industrial distillation of zinc at Zawar was established, direct alloying of pure metallic zinc and metallic copper became standard, producing high-grade brass with uniform stoichiometry ($60 - 70\%\ \mathrm{Cu}$, $30 - 40\%\ \mathrm{Zn}$).

Summary of Major Alloys

Alloy Name	Composition	Chemical Phase / Characteristics	Historical Applications
Wootz Steel	$\mathrm{Fe} + (1.0 - 1.6\%)\ \mathrm{C}$	Ultra-high carbon matrix with $\mathrm{Fe_3C}$ cementite bands	Damascus blades, high-strength tools
Bronze	$\mathrm{Cu} + (8 - 25\%)\ \mathrm{Sn}$	α or β-phase copper-tin solid solution	Chola icons, mirrors, vessels, weapons
Brass	$\mathrm{Cu} + (20 - 40\%)\ \mathrm{Zn}$	Face-centered cubic α-brass matrix	Utensils, coinage, decorative artifacts

Durability and Global Export of Indian Metals

The quality of ancient Indian metals earned them an international reputation. Indian iron and steel were highly sought after across West Asia, North Africa, and Europe. During the Roman Empire, high-quality steel exported from India (*Ferrum Indicum*) was recognized as superior to European iron for armaments.

The export of wootz steel ingots from ports like Barygaza (Bharuch) and Muchiri (Muziris) supplied Mediterranean and Middle Eastern markets for over a millennium. Syrian blacksmiths used Indian wootz ingots to forge Damascus swords, celebrated for their toughness and mechanical performance.

4. Fibre, Cloth, Paper, Ink and Dyeing Chemistry of Ancient India

The development of textiles, paper, inks, and synthetic mordant dyeing chemistry in ancient India highlights a sophisticated knowledge of natural organic compounds, surface chemistry, and material processing.

Fibres and Textile Technology

Ancient India was a global center for natural fibre processing, spinning, and weaving. Chemical analysis of textile remains reveals mastery over both plant-derived cellulosic polymers and animal-derived proteinaceous macromolecular materials.

Cotton

India is the birthplace of cultivated cotton (*Gossypium arboreum* and *Gossypium herbaceum*). Excavations at Harappan sites like Mohenjo-daro have yielded preserved cotton cloth fragments attached to silver artifacts, dating back to 3000 BCE.

Cotton consists of nearly pure **cellulose**, a natural linear polysaccharide polymer of D-glucose units connected by β-$(1 \rightarrow 4)$-glycosidic bonds:

$$(\mathrm{C_6H_{10}O_5})_n$$

The chemical structure of cotton cellulose contains multiple free hydroxyl ($-\mathrm{OH}$) groups per monomer unit:

These hydroxyl groups form extensive intra- and intermolecular hydrogen bonds, giving cotton high tensile strength, water absorption capacity, and reactive sites for chemical dye bonding.

Silk, Wool, and Flax

* **Silk:** Harvested from wild silkworms (*Tasar*, *Muga*, and *Eri*) and cultivated *Bombyx mori*, silk is a proteinaceous fibre consisting of **fibroin** coated with **sericin**. Fibroin is composed of repeating amino acid residues, primarily glycine (Gly), alanine (Ala), and serine (Ser):

$$-[\mathrm{Gly\text{-}Ala\text{-}Gly\text{-}Ser\text{-}Gly\text{-}Ala}]-_n$$

The antiparallel β-pleated sheet secondary structure of fibroin imparts high mechanical strength, elasticity, and lustre.

* **Wool:** Obtained from sheep and goats (e.g., Pashmina), wool consists of **keratin**, a fibrous protein rich in the sulfur-containing amino acid cysteine ($\mathrm{C_3H_7NO_2S}$). Interchain disulfide bonds ($-\mathrm{S-S}-$) give wool elasticity and heat insulation properties:

$$\mathrm{R\text{-}CH_2–S–S–CH_2\text{-}R}$$

* **Flax:** Derived from *Linum usitatissimum*, flax yields linen, a cellulosic bast fibre containing cellulose bound with lignin and hemicellulose.

Paper Making

Before the introduction of paper, written records were maintained on palm leaves (*Borassus flabellifer*) and the inner bark of the Himalayan birch tree (*Betula utilis*, known as *Bhurjapatra*). The bark was prepared by boiling, drying, and oiling to preserve its flexibility.

Paper manufacturing in India incorporated native organic pulping techniques using plant fibers, old rags, and hemp (*Cannabis sativa*). The raw materials were crushed, soaked in alkaline water (prepared from wood ash containing potassium carbonate, $\mathrm{K_2CO_3}$), and macerated to isolate insoluble cellulose fibers.

The cellulosic pulp was suspended in water and caught on fine mesh bamboo screens to form uniform sheets. Sizing agents were applied to reduce porous ink absorption:

* Rice starch or wheat starch (polysaccharides of amylose and amylopectin) served to bind surface fibers. * Alum ($\mathrm{KAl(SO_4)_2 \cdot 12H_2O}$) was added as a sizing fixative to precipitate starch onto the fibers and make the paper resistant to moisture and insect degradation.

Ink Chemistry

Ancient Indian scribes formulated durable black and colored inks for writing on palm leaves, birch bark, and paper.

Black Inks

Black ink (*Masi*) was prepared by collecting pure amorphous carbon (lampblack or soot) generated by the incomplete combustion of vegetable oils (such as mustard oil or sesame oil) or resinous woods:

$$\mathrm{Hydrocarbon\ Fuel + O_2\ (deficient) \rightarrow C(s,\ amorphous\ lampblack) + H_2O(g)}$$

The fine soot particles were thoroughly ground and suspended in a liquid medium containing a natural water-soluble polymer binder, such as *Gum Arabic* (extracted from *Acacia nilotica* tree sap, composed of complex polysaccharides) or plant mucilage. The binder prevented sedimentation of hydrophobic carbon particles by forming a protective colloidal envelope:

$$\mathrm{Carbon\ particles + Gum\ Arabic\ (binder) + Water \rightarrow Stable\ Colloidal\ Ink\ Suspension}$$

Copperas or green vitriol (ferrous sulfate, $\mathrm{FeSO_4 \cdot 7H_2O}$) combined with oak galls or plant tannins was also used to synthesize **iron-gall ink**. The reaction of tannic acid ($\mathrm{C_{76}H_{52}O_{46}}$) with ferrous ions forms a soluble ferrous tannate complex, which oxidizes upon atmospheric exposure to yield an insoluble black ferric tannate pigment:

$$\mathrm{Fe^{2+}(aq) + Tannic\ Acid \rightarrow Fe(II)\text{-}Tannate\ Complex\ (Soluble)}$$

$$\mathrm{4Fe(II)\text{-}Tannate + O_2(g) + 2H_2O(l) \rightarrow 4Fe(III)\text{-}Tannate\ Complex\ (Insoluble\ Black\ Pigment)}$$

Coloured Inks

Coloured inks were formulated by levigating insoluble inorganic minerals or organic extracts with binder solutions:

* **Red Ink:** Formulated from vermilion/cinnabar (mercuric sulfide, HgS) or red lead (minium, $\mathrm{Pb_3O_4}$). * **Yellow Ink:** Prepared using orpiment (arsenic trisulfide, $\mathrm{As_2S_3}$) or turmeric extracts. * **White Ink:** Formulated using finely ground chalk ($\mathrm{CaCO_3}$) or kaolin clay.

Dyeing Chemistry and Natural Dyes

Ancient Indian artisans mastered the chemical functionalization of natural dyes and their fixation onto textile fibers using inorganic mordants.

Natural Dyes

1. **Indigo (*Indigofera tinctoria*):**

Indigo is a dark blue vat dye extracted from the leaves of *Indigofera tinctoria*. The leaves contain the precursor glycoside **indican**. Fermentation converts indican via enzymatic hydrolysis to **indoxyl**:

$$\text{Indican} + \mathrm{H_2O} \xrightarrow{\text{glycosidase}} \text{Indoxyl} + \text{D-Glucose}$$

Indoxyl is water-soluble in alkaline solution. Upon exposure to atmospheric oxygen, two molecules of indoxyl undergo oxidative dimerization to form insoluble **indigotin** ($\mathrm{C_{16}H_{10}N_2O_2}$):

$$2\,\text{Indoxyl} + \mathrm{O_2(g)} \rightarrow \text{Indigotin (Blue Pigment)} + 2\mathrm{H_2O(l)}$$

Because indigotin is insoluble in water, it cannot directly bind to cotton fibers. Ancient dyers used a vat reduction process to convert indigotin into its soluble, reduced **leuco-indigo** form using alkaline fermentation (using wood ash containing $\mathrm{K_2CO_3}$ and reducing sugars):

$$\text{Indigotin (Insoluble)} + 2\mathrm{e^-} + 2\mathrm{H^+} \xrightarrow{\text{reduction}} \text{Leuco-indigo (Soluble)}$$

The cloth was immersed in the clear leuco-indigo solution. Upon removing the cloth and exposing it to air, oxidation regenerated the insoluble indigotin inside the fiber matrix:

$$\text{Leuco-indigo} + \frac{1}{2}\mathrm{O_2(g)} \xrightarrow{\text{air oxidation}} \text{Indigotin (Trapped in Fibres)} + \mathrm{H_2O(l)}$$

1. **Madder (*Rubia tinctorum* / *Rubia cordifolia* - Manjistha):**

The roots of *Rubia cordifolia* yield **alizarin** (1,2-dihydroxyanthraquinone, $\mathrm{C_{14}H_8O_4}$). Alizarin is a mordant dye that forms vivid red coordination complexes with metallic cations bound to cellulose or silk fibers.

1. **Turmeric (*Curcuma longa*):**

Yields **curcumin** ($\mathrm{C_{21}H_{20}O_6}$), a bright yellow polyphenolic compound that acts as a direct dye for silk and cotton.

1. **Lac Dye (*Kerria lacca*):**

Derived from the resinous secretion of the lac insect, containing anthraquinonoid derivatives known as **laccaic acids**, which produce deep crimson and scarlet shades on silk and wool.

1. **Saffron (*Crocus sativus*):**

Yields **crocin** ($\mathrm{C_{44}H_{64}O_{24}}$), a water-soluble carotenoid glycoside producing rich golden-yellow hues.

Mordants and Fixation Chemistry

Direct natural dyes like alizarin often lack affinity for cotton cellulose, leading to poor color fastness against washing and sunlight. Ancient Indian dyers resolved this by using **mordants**—polyvalent metal salts that act as chemical bridges between the fiber polymer and the dye molecule.

Commonly used ancient mordants included: * **Potassium Alum:** $\mathrm{KAl(SO_4)_2 \cdot 12H_2O}$ (prepared from alum-bearing shales). * **Green Vitriol / Iron Salts:** $\mathrm{FeSO_4 \cdot 7H_2O}$ (prepared by rusting iron in vinegar or fermented liquors).

The chemical mechanism of mordanting operates through coordinate covalent complexation:

1. **Hydrolysis of Metal Ions:** When alum dissolves in neutral water, aluminum ions undergo hydrolysis to form insoluble aluminum hydroxide cations bound to cellulosic hydroxyl groups:

$$\mathrm{Al^{3+}(aq) + 3H_2O(l) \rightleftharpoons Al(OH)_3(s) + 3H^+(aq)}$$

$$\mathrm{Cellulose\text{–}OH + Al(OH)_2^+ \rightarrow Cellulose\text{–}O\text{–}Al(OH)_2 + H^+}$$

1. **Chelation with Dye Molecule:** The fiber-bound metal ion acts as a Lewis acid center, coordinating with the electron-rich orthohydroxy-carbonyl groups of the dye molecule (such as alizarin) to form a stable six-membered chelate ring called a **lake**:

$$\mathrm{Cellulose\text{–}O\text{–}Al^{2+} + Alizarin^{2-} \rightarrow [Cellulose\text{–}O\text{–}Al\text{–}Alizarinate\ Complex]\ (Insoluble\ Color\ Lake)}$$

The formation of this insoluble Coordination Complex firmly anchors the dye within the interior structure of the fiber, preventing it from leaching out during washing. The use of iron mordants ($\mathrm{Fe^{2+}/Fe^{3+}}$) transformed red alizarin dye into dark violet, brown, or black complexes, allowing dyers to obtain a wide spectrum of light-fast colors from a single dye source.

Export of Indian Textiles

The mastery of mordant dyeing gave ancient Indian textiles exceptional color fastness, vibrant shades, and intricate patterns. Products such as *calico* (plain-woven unbleached cotton named after Calicut) and *chintz* (printed or painted glazed cotton textiles) became major commodities in international commerce.

Indian dyed textiles were exported in vast quantities across the Indian Ocean network to Southeast Asia, East Africa, the Persian Gulf, the Red Sea, and later to Europe via global trading companies. The dominance of Indian textiles in world trade persisted for over two millennia, driven by the chemical superiority of Indian mordant-dyeing techniques.

5. Discoveries and Inventions and Development of Chemistry during the Industrial Revolution

Discoveries and Inventions in Ancient Indian Chemistry

Ancient Indian chemical technology produced significant innovations across metallurgical, pharmaceutical, and consumer processes. The integration of chemical empirical knowledge with practical apparatus design led to several early technological achievements.

Distillation and Fermentation

The art of distillation and fermentation in ancient India was widely practiced for the manufacture of alcoholic beverages, medicated spirits, and essential oils. Early texts such as the *Charaka Samhita* and *Sushruta Samhita* describe various fermented preparations known as *Asavas* and *Arishtas*. Fermentation involved introducing plant juices, decoctions, or

herbal extracts containing natural sugars into sealed earthen vessels along with natural fermenting agents such as *Dhataki* flowers (*Woodfordia fruticosa*).

During fermentation, complex carbohydrates and disaccharides undergo enzymatic hydrolysis into simple hexose sugars, which subsequently break down into ethanol and carbon dioxide:

$$\mathrm{C_6H_{12}O_6}(aq) \xrightarrow{\text{zymase}} 2\mathrm{C_2H_5OH}(aq) + 2\mathrm{CO_2}(g)$$

To purify alcohol or extract aromatic compounds for perfumes (*Attar*), specialised distillation apparatus were invented. The *Vaka-yantra* and *Tiryakpatana-yantra* (transverse distillation apparatus) were designed using connected clay retorts and receivers. Volatile essential oils and alcoholic vapors were vaporized in a heated lower vessel, carried through an inclined tube, and condensed in a water-cooled receiver:

$$\text{Liquids (Mixture)} + \Delta \rightarrow \text{Vapor Phase} \xrightarrow{\text{Condenser}} \text{Purified Distillate}$$

This technique allowed the extraction of concentrated fragrant essences from rose, sandalwood, and jasmine, laying the foundation for a flourishing perfume industry.

Preparation of Zinc by Distillation

One of the most remarkable achievements of ancient Indian chemistry was the industrial-scale extraction of metallic zinc by downward distillation (*Tiryakpatana*). Zinc has a low boiling point of $907^\circ\mathrm{C}$, whereas the temperature required to reduce zinc oxide with carbon is above $1000^\circ\mathrm{C}$. In an open furnace, metallic zinc vaporizes immediately upon reduction and oxidizes back to zinc oxide in air. Ancient Indian metallurgists at Zawar in Rajasthan solved this thermodynamic challenge by devising a specialized closed-container downward distillation technique.

The principal zinc ore used was calamine (smithsonite, $\mathrm{ZnCO_3}$) or sphalerite (ZnS). Calamine was first calcined by heating to decompose the carbonate into zinc oxide:

$$\mathrm{ZnCO_3}(s) \xrightarrow{\Delta} \mathrm{ZnO}(s) + \mathrm{CO_2}(g)$$

The resulting zinc oxide was mixed with organic reducing agents such as charcoal, cow dung, and natural resins, along with fluxing agents. This mixture was placed inside brinjal-shaped clay retorts (*vrentaka-musha*). A long clay tube was attached to the open end of the retort. The retorts were inverted over a perforated hearth, with the clay tubes extending downwards into a lower cooling chamber containing water vessels.

Upon heating the upper section of the retort in a furnace, the carbon reduces zinc oxide inside the closed anaerobic environment:

$$\mathrm{ZnO}(s) + \mathrm{C}(s) \xrightarrow{\Delta > 1000^\circ\mathrm{C}} \mathrm{Zn}(g) + \mathrm{CO}(g)$$

The generated zinc vapor travels downwards through the clay tube into the lower cool zone, where it condenses directly into metallic zinc pellets before it can re-oxidize:

$$\mathrm{Zn}(g) \xrightarrow{\text{Cooling}} \mathrm{Zn}(s)$$

This downward distillation system represents the earliest historical evidence of pure metallic zinc isolation on an industrial scale, centuries before similar processes were established in Europe.

Manufacture of Soap, Glass, and Pottery

The manufacture of cleansing agents, glass, and ceramic items formed a major component of ancient Indian chemical arts. Early soaps (*Phenaka*) were prepared by boiling plant ashes rich in soluble alkali carbonates with vegetable oils and

animal fats. Alkali carbonates such as potassium carbonate ($\mathrm{K_2CO_3}$) derived from wood ash, or natural sodium carbonate deposits (*Sajjikshara*, $\mathrm{Na_2CO_3}$), were treated with slaked lime ($\mathrm{Ca(OH)_2}$) to form caustic alkalis:

$$\mathrm{Na_2CO_3(aq) + Ca(OH)_2(aq) \rightarrow 2NaOH(aq) + CaCO_3(s)}$$

The resulting caustic solution was reacted with triacylglycerols (triglycerides present in oils) to yield soap (fatty acid salts) and glycerol through saponification:

$$\mathrm{C_3H_5(OOCR)_3 + 3NaOH \rightarrow 3RCOONa + C_3H_5(OH)_3}$$

Glass manufacture (*Kacha*) in ancient India involved fusing silica ($\mathrm{SiO_2}$, sand) with alkali fluxes such as plant ash or *Sajjikshara* to lower the melting temperature of quartz. Additives such as lime (CaO) were incorporated to impart chemical durability against water, yielding soda-lime-silicate glass:

$$\mathrm{SiO_2(s) + Na_2CO_3(s) + CaCO_3(s) \xrightarrow{\Delta} Na_2O \cdot CaO \cdot 4SiO_2(s) + 2CO_2(g)}$$

Specific metallic oxides were added to impart distinct colors to glass beads and vessels, such as copper oxides for blue or red hues:

$$\mathrm{CuO + Silicate\ Matrix \rightarrow Blue\ Glass}$$

Pottery technology included high-temperature firing of refined clays containing aluminosilicates ($\mathrm{Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O}$). Glazed pottery was produced by applying a surface layer of low-melting silicate slurry containing iron or copper oxides, forming an impervious decorative vitreous coating during kiln firing.

Use of Gunpowder and Its Applications

The knowledge of explosive mixtures containing saltpetre (potassium nitrate, $\mathrm{KNO_3}$), charcoal (C), and sulphur (S) developed in India and was recorded in texts such as the *Sukraniti*. Saltpetre was harvested from natural soil efflorescences rich in organic nitrifying bacterial deposits.

Gunpowder functions as a pyrotechnic mixture where potassium nitrate acts as the oxidizing agent, while charcoal and sulphur serve as fuels. The overall chemical reaction occurring during rapid deflagration can be expressed as:

$$\mathrm{10KNO_3(s) + 3S(s) + 8C(s) \rightarrow 5K_2CO_3(s) + 3K_2SO_4(s) + 6CO_2(g) + 5N_2(g)}$$

The rapid oxidation of carbon to carbon dioxide and nitrogen gas produces a massive volumetric expansion and liberation of thermal energy:

$$\Delta H < 0$$

Gunpowder was utilized in pyrotechnics, military incendiary devices, rock blasting, and early artillery applications across medieval Indian kingdoms.

Impact on Indian Society, Trade, and Industry

These chemical technologies significantly influenced ancient and medieval Indian economic history. High-purity zinc, perfumes, glass beads, and refined textiles were traded extensively across Asia, the Mediterranean, and Africa. The production of high-carbon steel (*Wootz steel*) and zinc metal established India as a major exporter of advanced material goods. Domestic industries in soap, glass, dyes, and ceramics supported urban centers, created specialized craftsman guilds (*Srenis*), and generated tax revenues for states.

Transfer of Indian Chemical Knowledge to the Arab World and Europe

During the 8th to 12th centuries CE, the Abbasid Caliphate in Baghdad instituted extensive programs to translate Indian scientific and medical texts into Arabic. Works of Charaka, Sushruta, and Vagbhata were translated into Arabic under scholars such as Yahya ibn Khalid. Chemical and alchemical manuscripts detailing apparatus construction, metallurgical refining, and mercurial processing reached Islamic scholars including Jabir ibn Hayyan (Geber) and Al-Razi (Rhazes).

Al-Razi incorporated Indian chemical classifications and distillation techniques into his text *Kitab al-Asrar* (*Book of Secrets*). Later, during the 12th century Latin translation movement in Spain and Italy, these Arabic texts were translated into Latin, transmitting Indian chemical techniques—such as downward distillation and saltpetre purification—into medieval European laboratory practices.

Development of Chemistry During the Industrial Revolution

Shift from Craft Chemistry to Industrial Chemistry

The 18th and 19th centuries marked a major transition from small-scale empirical craft chemistry to large-scale, continuous industrial chemical processing. This transformation was driven by the integration of stoichiometry, chemical thermodynamics, mechanical power, and mass transport design. Industries shifted from batch production in small clay or glass vessels to high-volume manufacturing using specialized industrial reactors, lead chambers, and furnaces.

Large-Scale Manufacture of Sulphuric Acid: The Lead Chamber Process

Sulphuric acid ($\mathrm{H_2SO_4}$) was the fundamental chemical building block of the Industrial Revolution, required for processing textiles, metals, alkalis, and fertilizers. The Lead Chamber Process, introduced by John Roebuck in 1746 and later improved by Gay-Lussac and Glover, enabled mass production.

Sulphur or iron pyrites ($\mathrm{FeS_2}$) were burned in air to produce sulphur dioxide gas:

$$\mathrm{S}(s) + \mathrm{O_2}(g) \rightarrow \mathrm{SO_2}(g)$$

$$4\mathrm{FeS_2}(s) + 11\mathrm{O_2}(g) \rightarrow 2\mathrm{Fe_2O_3}(s) + 8\mathrm{SO_2}(g)$$

The sulphur dioxide gas was introduced into large lead-lined chambers along with steam, air, and nitrogen oxide gases (NO and $\mathrm{NO_2}$) which acted as homogeneous catalysts. Nitrogen oxide reacts with atmospheric oxygen to form nitrogen dioxide:

$$2\mathrm{NO}(g) + \mathrm{O_2}(g) \rightarrow 2\mathrm{NO_2}(g)$$

Nitrogen dioxide oxidizes sulphur dioxide in the presence of water vapor to form sulphuric acid, regenerating nitric oxide:

$$\mathrm{SO_2}(g) + \mathrm{NO_2}(g) + \mathrm{H_2O}(l) \rightarrow \mathrm{H_2SO_4}(aq) + \mathrm{NO}(g)$$

The net overall reaction taking place within the lead chambers is:

$$2\mathrm{SO_2}(g) + \mathrm{O_2}(g) + 2\mathrm{H_2O}(l) \xrightarrow{\mathrm{NO}_x} 2\mathrm{H_2SO_4}(aq)$$

This process supplied the high volumes of concentrated acid needed for emerging industrial applications.

Manufacture of Alkalis: The Leblanc Process and the Solvay Process

Industrial growth created demand for sodium carbonate ($\mathrm{Na_2CO_3}$, soda ash), used in glassmaking, soap manufacture, and textile processing.

The Leblanc Process

Patented by Nicolas Leblanc in 1791, the Leblanc process was the first commercial method for synthesizing soda ash from sodium chloride.

In the first stage, salt cake (sodium sulphate) was produced by heating common salt with concentrated sulphuric acid:

$$2\mathrm{NaCl}(s) + \mathrm{H_2SO_4}(l) \rightarrow \mathrm{Na_2SO_4}(s) + 2\mathrm{HCl}(g)$$

In the second stage, the sodium sulphate was roasted in a rotary kiln with coal (carbon) and limestone (calcium carbonate) to form "black ash":

$$\mathrm{Na_2SO_4}(s) + 2\mathrm{C}(s) \rightarrow \mathrm{Na_2S}(s) + 2\mathrm{CO_2}(g)$$

$$\mathrm{Na_2S}(s) + \mathrm{CaCO_3}(s) \rightarrow \mathrm{Na_2CO_3}(s) + \mathrm{CaS}(s)$$

Combining these two red-hot reduction and exchange reactions gives the net black ash reaction equation:

$$\mathrm{Na_2SO_4}(s) + 2\mathrm{C}(s) + \mathrm{CaCO_3}(s) \rightarrow \mathrm{Na_2CO_3}(s) + \mathrm{CaS}(s) + 2\mathrm{CO_2}(g)$$

The soluble sodium carbonate was extracted from the insoluble calcium sulphide waste by lixiviation with water and purified by crystallization.

The Solvay Process

In the 1860s, Ernest Solvay developed a more efficient, less polluting continuous chemical process known as the Ammonia-Soda or Solvay process.

Brine (saturated sodium chloride solution) is saturated with gaseous ammonia ($\mathrm{NH_3}$) and then treated with carbon dioxide ($\mathrm{CO_2}$) under pressure. Carbon dioxide reacts with aqueous ammonia to form ammonium bicarbonate:

$$\mathrm{NH_3}(aq) + \mathrm{CO_2}(g) + \mathrm{H_2O}(l) \rightarrow \mathrm{NH_4HCO_3}(aq)$$

Ammonium bicarbonate reacts with sodium chloride to precipitate sparingly soluble sodium bicarbonate:

$$\mathrm{NaCl}(aq) + \mathrm{NH_4HCO_3}(aq) \rightarrow \mathrm{NaHCO_3}(s)\downarrow + \mathrm{NH_4Cl}(aq)$$

The overall precipitation equation is:

$$\mathrm{NaCl}(aq) + \mathrm{NH_3}(aq) + \mathrm{CO_2}(g) + \mathrm{H_2O}(l) \rightarrow \mathrm{NaHCO_3}(s)\downarrow + \mathrm{NH_4Cl}(aq)$$

The precipitated sodium bicarbonate is filtered and calcined by heating to yield solid sodium carbonate:

$$2\mathrm{NaHCO_3}(s) \xrightarrow{\Delta} \mathrm{Na_2CO_3}(s) + \mathrm{H_2O}(g) + \mathrm{CO_2}(g)$$

Ammonia is recycled within a closed loop by treating ammonium chloride solution with slaked lime ($\mathrm{Ca(OH)_2}$), derived from burning limestone:

$$\mathrm{CaCO_3}(s) \xrightarrow{\Delta} \mathrm{CaO}(s) + \mathrm{CO_2}(g)$$

$$\mathrm{CaO}(s) + \mathrm{H_2O}(l) \rightarrow \mathrm{Ca(OH)_2}(aq)$$

$$2\mathrm{NH_4Cl}(aq) + \mathrm{Ca(OH)_2}(aq) \rightarrow 2\mathrm{NH_3}(g)\uparrow + \mathrm{CaCl_2}(aq) + 2\mathrm{H_2O}(l)$$

The Solvay process replaced the Leblanc process due to higher efficiency, lower thermal energy demand, and reduced environmental waste generation.

Chlorine and Bleaching Powder

The expansion of the cotton textile industry required rapid bleaching methods to replace traditional sun-bleaching. Chlorine gas was manufactured by oxidizing hydrogen chloride (a byproduct of the Leblanc process) with manganese dioxide (

MnO_2):

$$MnO_2(s) + 4HCl(aq) \rightarrow MnCl_2(aq) + Cl_2(g) + 2H_2O(l)$$

Because toxic chlorine gas was difficult to transport safely, Charles Tennant in 1799 developed solid bleaching powder by reacting gaseous chlorine with dry slaked lime:

$$Ca(OH)_2(s) + Cl_2(g) \rightarrow CaOCl_2(s) + H_2O(l)$$

Bleaching powder ($CaOCl_2$, calcium hypochlorite chloride) released hypochlorous acid ($HClO$) when dissolved in water in the presence of dilute acid, yielding active oxygen for oxidative decolorization of textile fibers:

$$CaOCl_2(s) + H_2SO_4(aq) \rightarrow CaSO_4(s) + HCl(aq) + HClO(aq)$$

$$HClO(aq) \rightarrow HCl(aq) + [O]$$

Growth of the Dye Industry: From Natural to Synthetic Dyes

Prior to the mid-19th century, dyes were derived exclusively from natural biological sources such as madder root (alizarin), indigo leaves, and cochineal insects. These extraction processes produced variable dye quality at high cost.

In 1856, William Henry Perkin revolutionized organic chemistry while attempting to synthesize the antimalarial drug quinine. By oxidizing crude aniline (containing *o*- and *p*-toluidine impurities derived from coal tar) with potassium dichromate ($K_2Cr_2O_7$) in sulphuric acid, he isolated a purple synthetic dye named Mauveine (aniline purple):

$$C_6H_5NH_2 + CH_3C_6H_4NH_2 \xrightarrow{K_2Cr_2O_7/H_2SO_4} \text{Mauveine (Chromophore Salts)}$$

This discovery triggered the creation of the synthetic dyestuff industry. It demonstrated that organic compounds could be produced at industrial scales from coal-tar waste byproducts, spurring synthetic production of alizarin (1869) and indigo (1897), and laying the groundwork for the modern pharmaceutical industry.

Development of Metallurgy: The Bessemer Process for Steel

The demand for strong structural materials for railways, steamships, and machinery led to advances in ferrous metallurgy. Pig iron produced in blast furnaces contained high carbon content ($3 - 5\%$), rendering it brittle.

In 1856, Henry Bessemer patented the Bessemer Process for mass-producing steel by decarbonizing pig iron. Air was blown through molten pig iron contained inside a pear-shaped refractory-lined vessel (the Bessemer converter). Oxygen in the air blast oxidizes impurities such as silicon, manganese, and carbon directly out of the molten metal:

$$Si(l) + O_2(g) \rightarrow SiO_2(s)$$

$$2Mn(l) + O_2(g) \rightarrow 2MnO(s)$$

Exothermic oxidation reactions generate sufficient heat to raise the temperature of the molten mass above the melting point of pure iron ($1538^\circ C$) without requiring external fuel:

$$\Delta H < 0$$

Carbon is oxidized to carbon monoxide, which burns at the mouth of the converter:

$$C(l) + \frac{1}{2}O_2(g) \rightarrow CO(g)$$

$$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$$

The silica and manganese oxides combine to form a slag:

$$\mathrm{MnO}(s) + \mathrm{SiO_2}(s) \rightarrow \mathrm{MnSiO_3}(l)$$

Once impurities were purged, controlled quantities of carbon and manganese (as spiegeleisen or ferromanganese) were added back into the melt to produce steel with precise carbon concentration ($0.1 - 1.5\%$). This reduced production costs and enabled modern infrastructure development.

Rise of the Chemical Industry and Its Impact on Society and Environment

The expansion of industrial chemistry altered economic and social structures. Mass-produced chemicals lowered the cost of soap, paper, textiles, glass, and steel, improving public sanitation and material living standards.

However, early chemical manufacturing produced severe environmental pollution:

1. The Leblanc process vented large volumes of gaseous hydrogen chloride (HCl) into the atmosphere, causing acid rain that devastated surrounding vegetation and infrastructure.
2. The solid waste residue from the Leblanc process—alkali waste containing calcium sulphide (CaS), known as "galligu"—accumulated in immense mounds. Moisture and atmospheric carbon dioxide reacted with these heaps to release toxic, hydrogen sulphide gas:

$$\mathrm{CaS}(s) + \mathrm{CO_2}(g) + \mathrm{H_2O}(l) \rightarrow \mathrm{CaCO_3}(s) + \mathrm{H_2S}(g)$$

1. Coal burning in blast furnaces, lead chamber plants, and dye factories released sulphur dioxide ($\mathrm{SO_2}$) and heavy metal residues into urban air and waterways.

These environmental crises prompted early regulatory legislation, such as Britain's Alkali Act of 1863, which mandated that factories absorb at least 95% of produced hydrogen chloride gas in water towers:

$$\mathrm{HCl}(g) + \mathrm{H_2O}(l) \rightarrow \mathrm{HCl}(aq)$$

This led to the commercial recovery of hydrochloric acid as a valuable feedstock for chlorine production. The evolution of industrial chemistry highlighted the need to balance chemical production efficiency with environmental monitoring and waste recovery technologies.

Questions

Short form first (2–3 marks), then long form (5–10 marks). **No Assam University question was found for this unit** — the questions below are syllabus-derived, marked as such.

Short-form questions

S1. [standard] What is Ayurvedic chemistry? Name two preparation methods used in Ayurveda.

Answer in brief: Section 2 — Ayurvedic chemistry is the preparation of medicines from plant, mineral and animal products; methods include decoctions (kwatha), powders (churna), fermented preparations (asava, arishta) and calcination (bhasma).

S2. [standard] What is alchemy (Rasashastra) in India? Name two of the eight metals of Rasashastra.

Answer in brief: Section 2 — Rasashastra is the ancient Indian alchemical tradition of preparing elixirs and transmuting base metals; its eight metals include gold, silver, copper, iron, tin, lead, zinc and mercury.

S3. [standard] Name two metals or alloys produced in ancient India and one use of each.

Answer in brief: Section 3 — wootz steel (swords and tools, exported to West Asia), bronze (images and vessels), zinc from Zawar (alloys and medicines).

S4. [standard] Name two natural dyes used in ancient India and the mordant used to fix them.

Answer in brief: Section 4 — indigo, madder, turmeric and lac; mordants such as alum or iron salts fix the dye on cotton and silk.

Long-form questions

L1. [standard] Describe old traditions of chemical sciences in India with examples.

What to write: Section 1 in full — Vedic references to metals, dyes and fermentation, the five elements and Kanada's atomic ideas, and chemical knowledge in the Arthashastra, Charaka Samhita and Sushruta Samhita.

L2. [standard] Write a note on Ayurvedic Chemistry and on chemistry in medical schools of ancient India.

What to write: Section 2 in full — doshas and drug preparations (kwatha, churna, asava, arishta, bhasma), Rasashastra and the eight metals, and the medical schools of Taxila and Nalanda with the Charaka and Sushruta Samhitas.

L3. [standard] Describe metal and metallurgy in ancient India, and fibre, cloth, paper, ink and dyeing chemistry of ancient India.

What to write: Sections 3–4 in full — iron, copper, zinc and wootz steel; cotton, silk and wool; paper and ink; natural dyes and mordants.

L4. [standard] Describe discoveries and inventions in ancient Indian chemistry in the context of state of art and impact, and development of chemistry during the industrial revolution.

What to write: Section 5 in full — distillation, fermentation, zinc, soap, glass and gunpowder and their impact, and the lead chamber, Leblanc, Solvay and Bessemer processes of the industrial revolution.

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.